

Anchoring Effects

The State of the Research of one of the
Most Robust Phenomena of Social Psychology

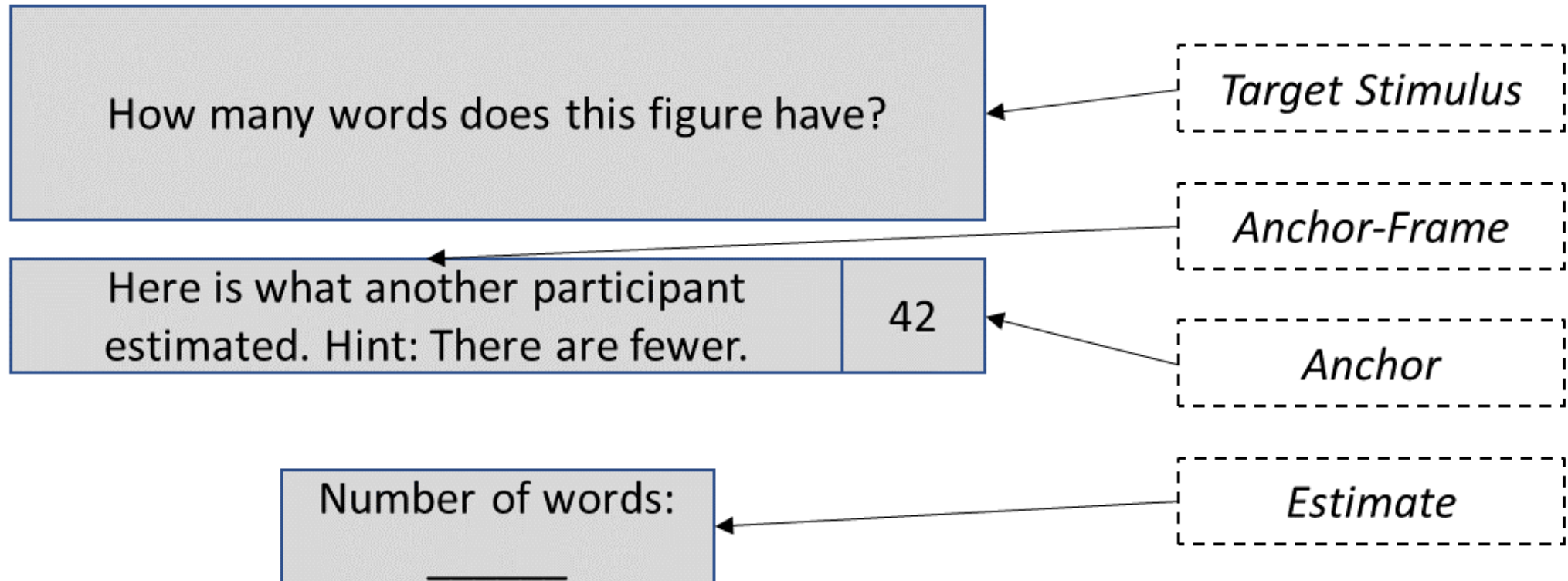


osf.io/tqx69

twitter.com/aufdroeseler

[lukas.roeseler\(at\)uni-bamberg.de](mailto:lukas.roeseler@uni-bamberg.de)

Anchoring Effects



Robustness of Anchoring

Effect size according to our meta-analysis:

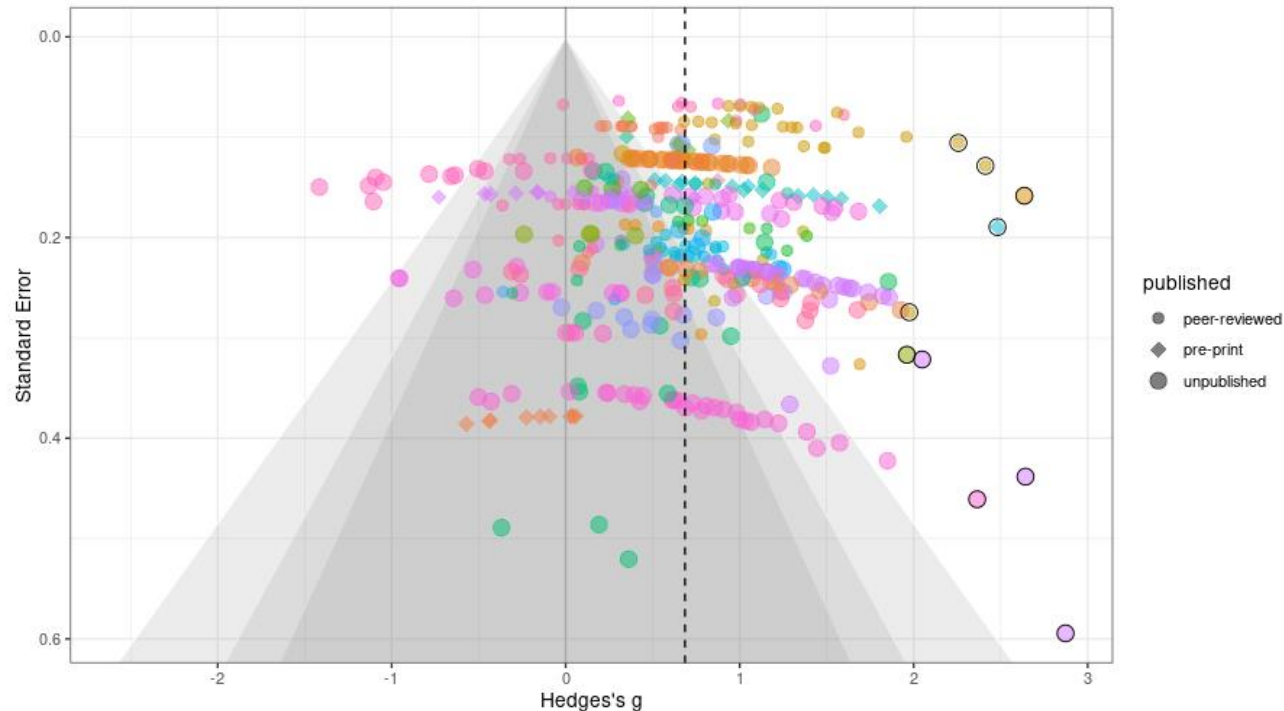
$g = 0.686$, 95% CI [0.586, 0.786],

95% PI [-0.241, 1.613],

$\sigma^2 = 0.22$,

$N_{total} = 18460$,

$k = 410$



One of the most reliable and robust results of experimental psychology
(Kahneman, 2012, p. 119)

Many labs effect sizes: $d = [1.0, 2.5]$
(Klein et al., 2014, p. 148)

One of the most consistent findings in psychology
(Cheek & Norem, 2019, p. 1)

1st, 3rd, 4th, and 6th largest effect sizes amongst average standardized mean difference [43 meta-analyses from social and cognitive psychology]
(Olsson-Collentine et al., 2020)

Large meta-analytical effect in legal decision-making, $d_{95\%CI} = [0.69, 1.12]$
(Bystranowski et al., 2021)

Meta-analytical effect size for WTP/WTa:
 $d_{95\%CI} = [0.41, 0.75]$
(Li et al., 2021)

Debates in Anchoring Research

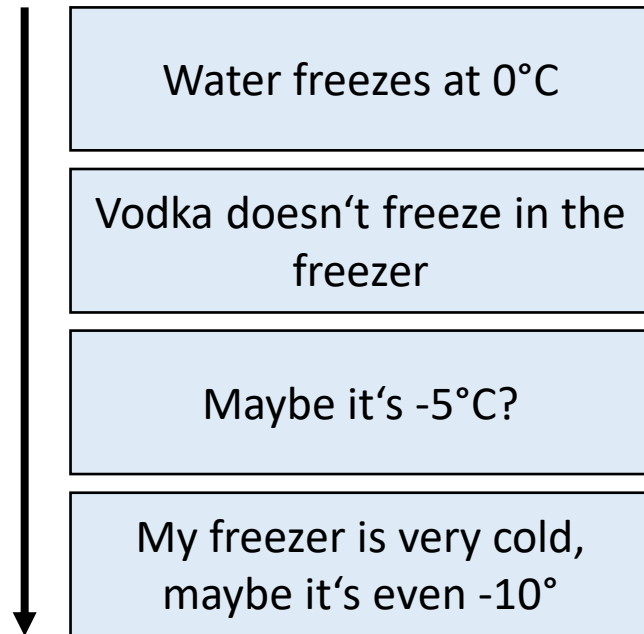
Moderator	Increases Anchoring	Decreases Anchoring
Ego Depletion	Banker et al., 2017	Haan & van Veldhuizen, 2015
Time-Pressure	Reitsma-van Rooijen & Daamen, 2006	Mussweiler & Englich, 2005
Expertise	Townson, 2019	e.g., Bystranowski et al., 2021
Openness	e.g., McElroy & Dowd, 2007	e.g., Eroglu & Croxton, 2010
Conscientiousness	e.g., Eroglu & Croxton, 2010	e.g., McElroy & Dowd, 2007
Agreeableness	Eroglu & Croxton, 2010	e.g., Furnham et al., 2012
Neuorticism	Jasper & Ortner, 2014	e.g., Perry & Sibley, 2013

Types of Anchors

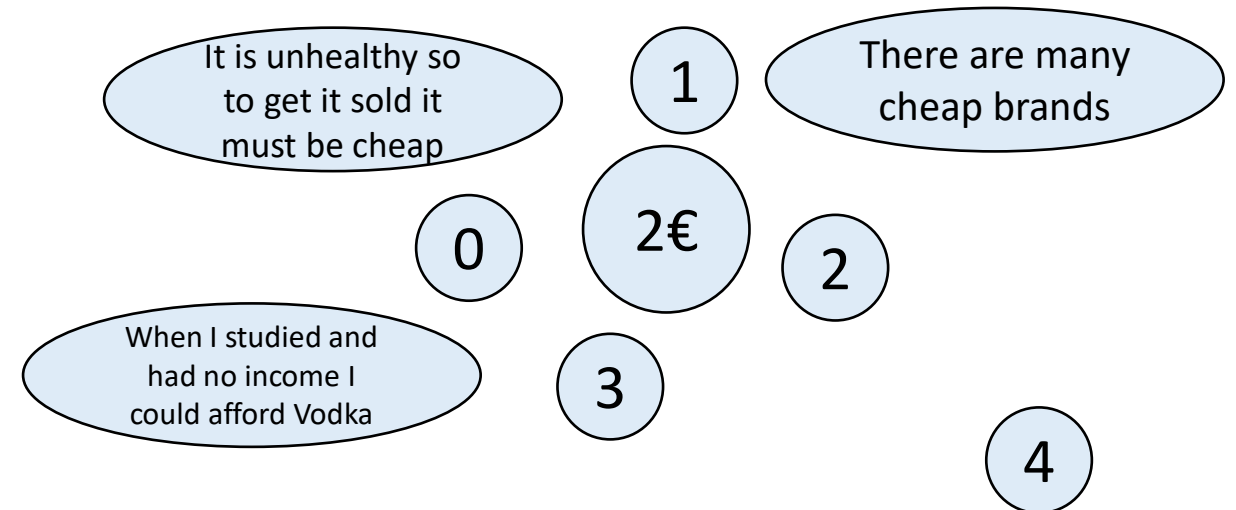
Self-generated versus experimenter-provided

Self-Generated versus Experimenter-Provided

What is the freezing temperature of Vodka?



Is the price of a bottle of Vodka more or less than 2€?



History of Anchoring Types

Anchoring-and-adjustment is prominently described by Kahneman and Tversky, 1974

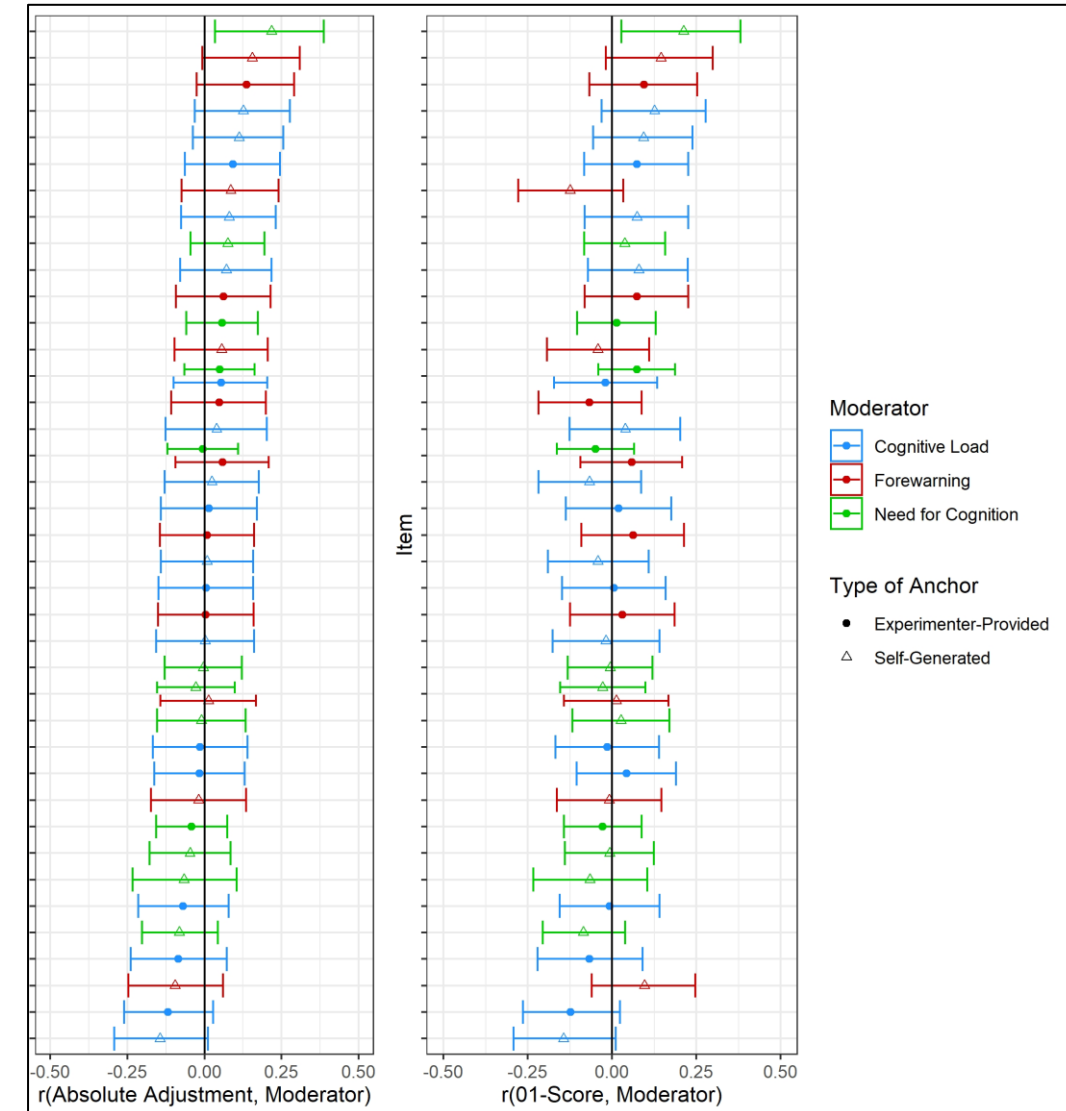
Mussweiler and English publish their Priming Model (1999a, 1999b)

Epley and Gilovich show that both models can coexist and each of them is valid for one type of anchoring, only (2001, 2004, 2005, 2006)

Simmons et al. (2010) and Chaxel (2014) show that both models are valid for both types

Harris et al. (2019) show that the SAM is wrong as key findings are not replicable → So how could these models coexist for 20 years?

Röseler et al. (2022) show that Epley and Gilovich's findings are not replicable either



Theories of Anchoring

Theories of Anchoring

Model	Exclusive Empirical Consistencies	Empirical Inconsistencies
Insufficient Adjustment	People make adjustment steps (Epley & Gilovich, 2001; Frech et al., 2020; Röseler et al., 2020)	Motivating accuracy does not work well (e.g., Simmons et al., 2010)
Selective Accessibility	Anchoring leads to consistent information being recognized faster (Mussweiler & Strack, 1999); positive correlation between anchoring effect strength and semantic similarity between the target and the anchor (Sailors and Heyman, 2019)	Evaluative judgment eliminates anchoring (Schweickhart, et al. 2020); Anchoring is scale specific (Frederick & Mochon, 2012); hypothesis-consistent testing is unplausible (Street & Kingstone, 2017); Different effects despite same semantics (Wong & Kwong, 2000)
Scale Distortion Theory	Anchoring is scale specific (Frederick & Mochon, 2012)	Self-generated anchors (e.g., Epley & Gilovich, 2001)
Incidental Environmental Anchoring	Numeric Priming (Critcher & Gilovich, 2008); Subliminal Anchoring (Mussweiler & Englich, 2005)	Contradictory Findings (Reitsma-van-Rooijen & Daamen, 2006)

Theories of Anchoring

Model	Exclusive Empirical Consistencies	Empirical Inconsistencies	Falsification attempts
Insufficient Adjustment	People make adjustment steps (Epley & Gilovich, 2001; Frech et al., 2020; Röseler et al., 2020)	Motivating accuracy does not work well (e.g., Simmons et al., 2010)	Time-pressure does not affect the strength of anchoring (Röseler et al., in preparation)
Selective Accessibility	Anchoring leads to consistent information being recognized faster (Mussweiler & Strack, 1999); positive correlation between anchoring effect strength and semantic similarity between the target and the anchor (Sailors and Heyman, 2019)	Evaluative judgment eliminates anchoring (Schweickhart, et al. 2020); Anchoring is scale specific (Frederick & Mochon, 2012); hypothesis-consistent testing is unplausible (Street & Kingstone, 2017); Different effects despite same semantics (Wong & Kwong, 2000)	Failures to replicate key result (Harris et al., 2019) low degree of empirical content (Röseler et al., 2019) “Bad” anchors don’t bias estimates (Schweickart et al., 2020) [but irrelevant anchors still have a small effect, Röseler et al., 2019]
Scale Distortion Theory	Anchoring is scale specific (Frederick & Mochon, 2012)	Self-generated anchors (e.g., Epley & Gilovich, 2001)	Previous anchors and judgments on the same scale have a negligible effect (Bahnik, 2021)
Incidental Environmental Anchoring	Numeric Priming (Critcher & Gilovich, 2008); Subliminal Anchoring (Mussweiler & Englich, 2005)	Contradictory Findings (Reitsma-van-Rooijen & Daamen, 2006)	Original results could not be replicated and are likely due to QRP (Shanks et al., 2020, Röseler et al., 2020)

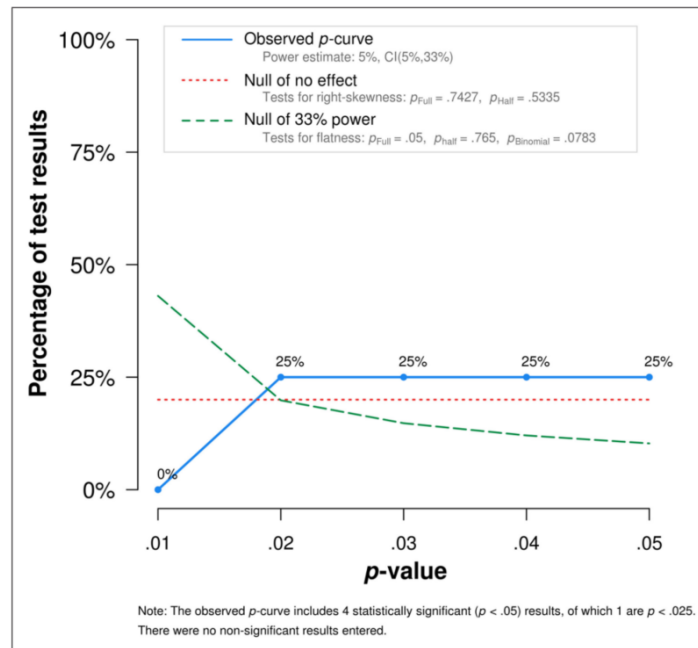
 Failed to replicate
 Serious contradiction

Prominence of Anchors

Focal/Supraliminal versus Incidental versus Subliminal

Incidental Anchoring

As predicted, those who saw Stan Fischer wearing a jersey number 94 estimated that his probability of sacking the quarterback in his next gam was higher ... than did those who saw him wearing number 54 ..., $t(258) = 2.07, p = .04, d = .26$.



Shanks et al., 2020, p. 11

Original studies

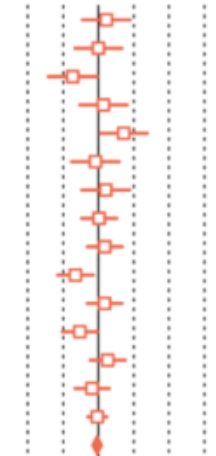
C & G (2008) Study 3 Restaurant
D-D & K (2015) Study 1 Restaurant
D-D & K (2015) Study 1b Restaurant
Average (random effects)



ES	Lower	Upper
0.40	0.12	0.69
0.54	0.02	1.06
0.98	0.22	1.74
0.49	0.25	0.72

Present studies: Incidental

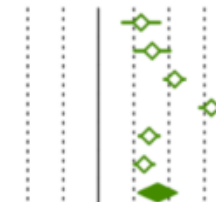
Study 1 Restaurant
Study 1 Hotel
Study 1 Membership
Study 1 VR glasses
Study 1 iPad
Study 1 Bar
Study 1 Holiday
Study 2 Restaurant
Study 2 Hotel
Study 2 Membership
Study 2 VR glasses
Study 2 iPad
Study 2 Bar
Study 2 Holiday
Study 3 Restaurant
Average (random effects)



0.11	-0.22	0.45
0.00	-0.33	0.33
-0.36	-0.71	-0.02
0.07	-0.26	0.41
0.36	0.02	0.69
-0.04	-0.38	0.30
0.10	-0.24	0.44
0.01	-0.24	0.26
0.09	-0.16	0.34
-0.32	-0.58	-0.07
0.09	-0.16	0.33
-0.26	-0.51	-0.02
0.13	-0.11	0.38
-0.09	-0.33	0.15
-0.01	-0.15	0.12
-0.02	-0.08	0.05

Present studies: Standard

Study 2 Population
Study 2 Length
Study 3 Population
Study 3 Birthrate
Study 3 Distance
Study 3 Year
Average (random effects)



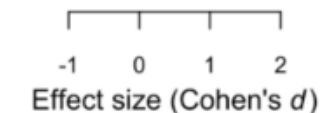
0.60	0.34	0.87
0.76	0.51	1.01
1.08	0.94	1.22
1.60	1.44	1.76
0.72	0.58	0.86
0.65	0.51	0.79
0.84	0.57	1.11

Random-effects meta-analyses

All effect sizes
Original + Incidental
Only Restaurant

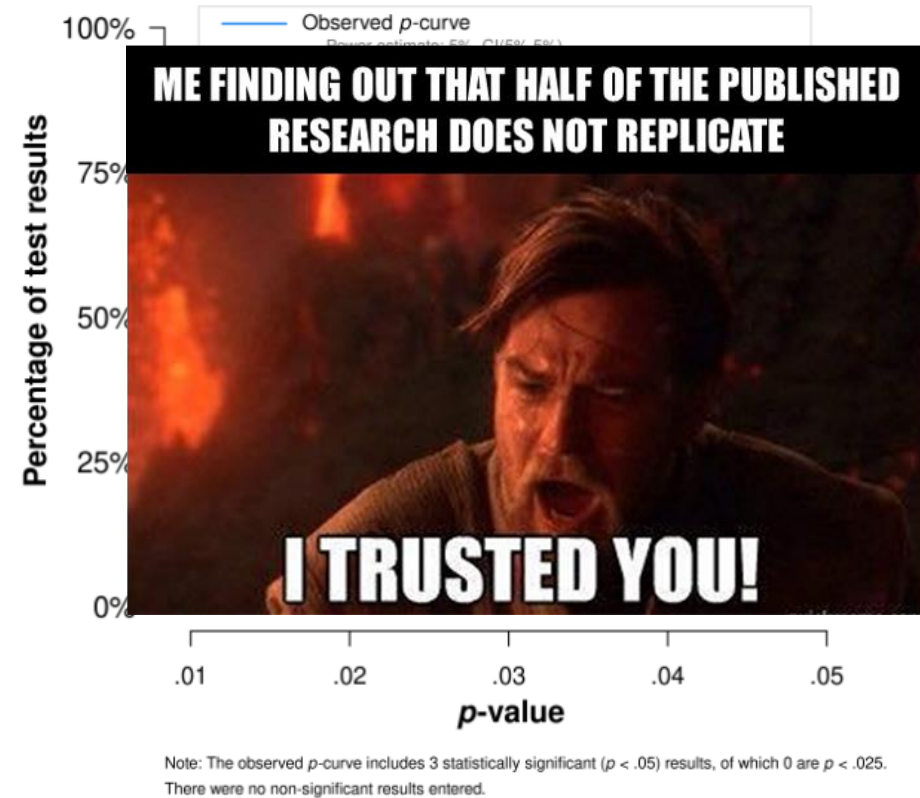


0.41	0.12	0.69
0.19	-0.04	0.42
0.22	0.00	0.45



Subliminal Anchoring

- Multiple studies by M&E + R&D that were published at the same time
- No published replications
- No further findings
- Incompatible paradigms (R&D set participants under time-pressure and M&E did not)
- It didn't replicate (Röseler et al., 2021)

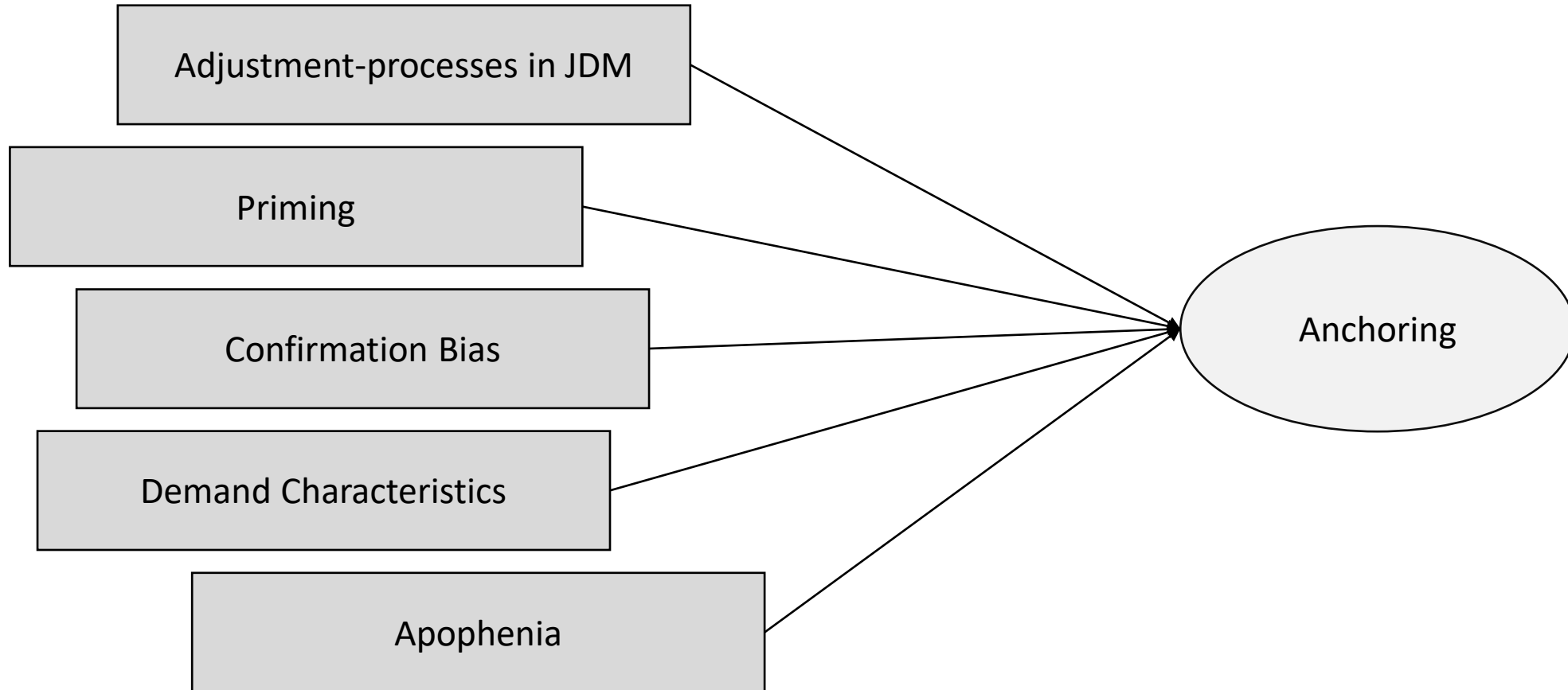


Study	Time pressure	Original N	Original d	Replication N	Replication d
M&E 2005, Studies 1 & 2	Absent	76	0.690	165	-0.148
R&D 2006a	Absent	[30, 32]	-0.28	79	0.226
R&D 2006b	Present	[30, 32]	0.867	91	0.104

The Role of Personality

The fruits of one decade of personality moderator research

Some of its Explanations are Traits



Moderators

- Big Five
- Cognitive Ability
- Cognitive Reflection
- Self-Control
- Narcissism
- Autistic tendencies

Correlations between Susceptibility to Anchoring Scores

Score	1	2	3	4	5	6	7	8	9	10
1 Quantity 50	1	.185**	.052	-.088	.054	.053	.136*	-.044	-.101	.061
2 Quantity 40	.185**	1	.087	.058	.091	.051	-.038	.029	.022	.024
3 Quantity 60	.052	.087	1	.111	.076	-.028	.028	-.048	-.026	.025
4 Age 69	-.088	.058	.111	1	-.078	.008	.093	-.002	-.007	.010
5 Age 53	.054	.091	.076	-.078	1	.054	-.114	-.110	.049	-.001
6 Age 36	.053	.051	-.028	.008	.054	1	.059	.020	.125	-.042
7 Age 30	.136*	-.038	.028	.093	-.114	.059	1	.057	.158*	-.002
8 WTP pen	-.044	.029	-.048	-.002	-.110	.020	.057	1	.101	-.007
9 WTP apple	-.101	.022	-.026	-.007	.049	.125	.158*	.101	1	-.003
10 WTP keyboard	.061	.024	.025	.010	-.001	-.042	-.002	-.007	-.003	1

Note. $N = 225$.

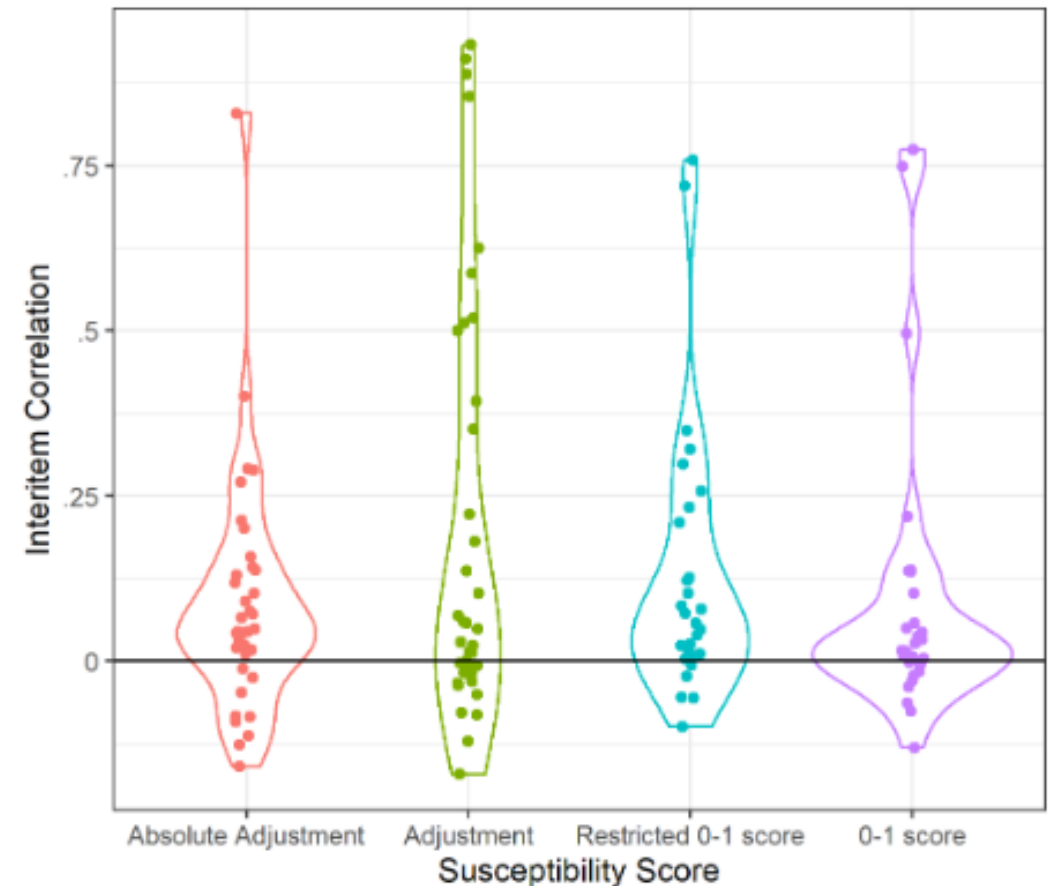
* $p < .05$ (two-tailed). ** $p < .01$ (two-tailed).

Röseler et al., 2019

People do not Care About Reliability

- We found out that after 12 years of personality research in anchoring, no one had reported reliability of adjustment scores (Röseler et al., 2019)
- Publishing our pre-print didn't change people's habits
- We aggregated all available anchoring datasets (currently 94) to compute reliability along different paradigms (Röseler et al., 2022)
- There is no evidence that you can measure susceptibility to anchoring reliably
- This doesn't stop people from publishing significant effects without reporting reliabilities (e.g., Cheek & Norem, 2022)

Violin Plots of Interitem Correlations by Susceptibility Score

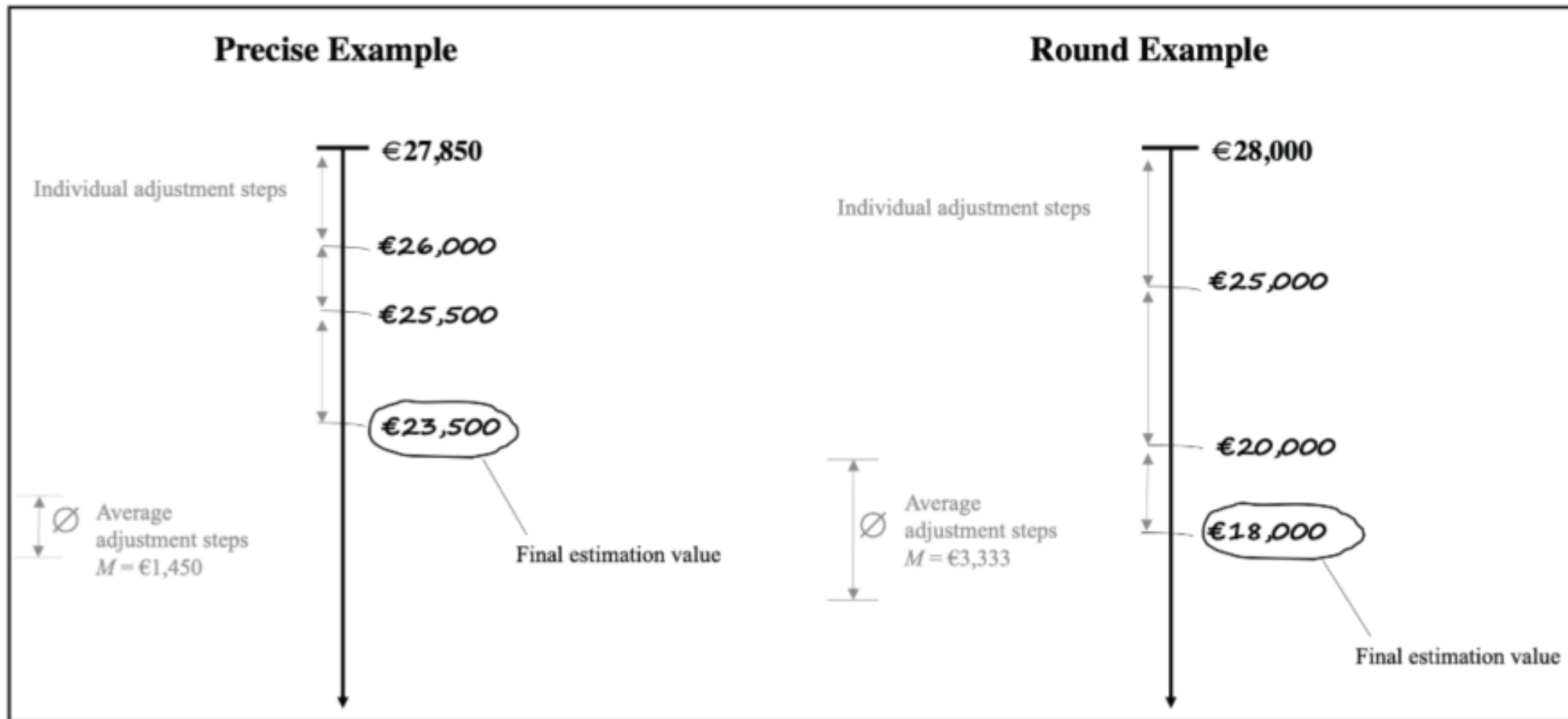


Moderators

What is left

Trustworthiness of Anchoring Findings

Frech et al., 2020, p. 47, Fig. 1

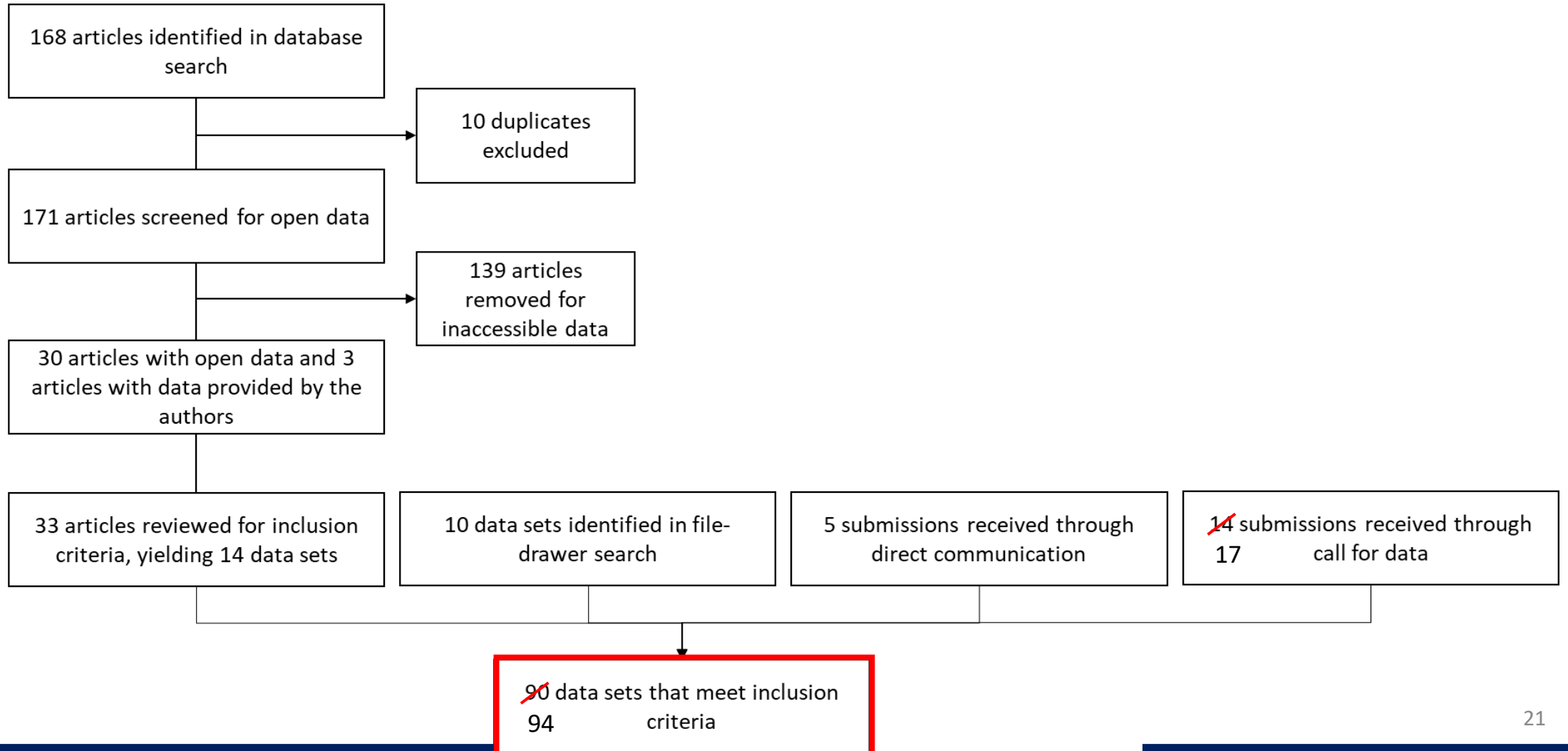


Unus pro omnibus

Centralizing Anchoring Research to Combat Challenges of Replicability

The OpAQ-dataset:

Aggregation of anchoring datasets (trial-level)



OpAQ: Unus pro omnibus

References	Studies	N	Items	Trials
55	94	20922	403	87490

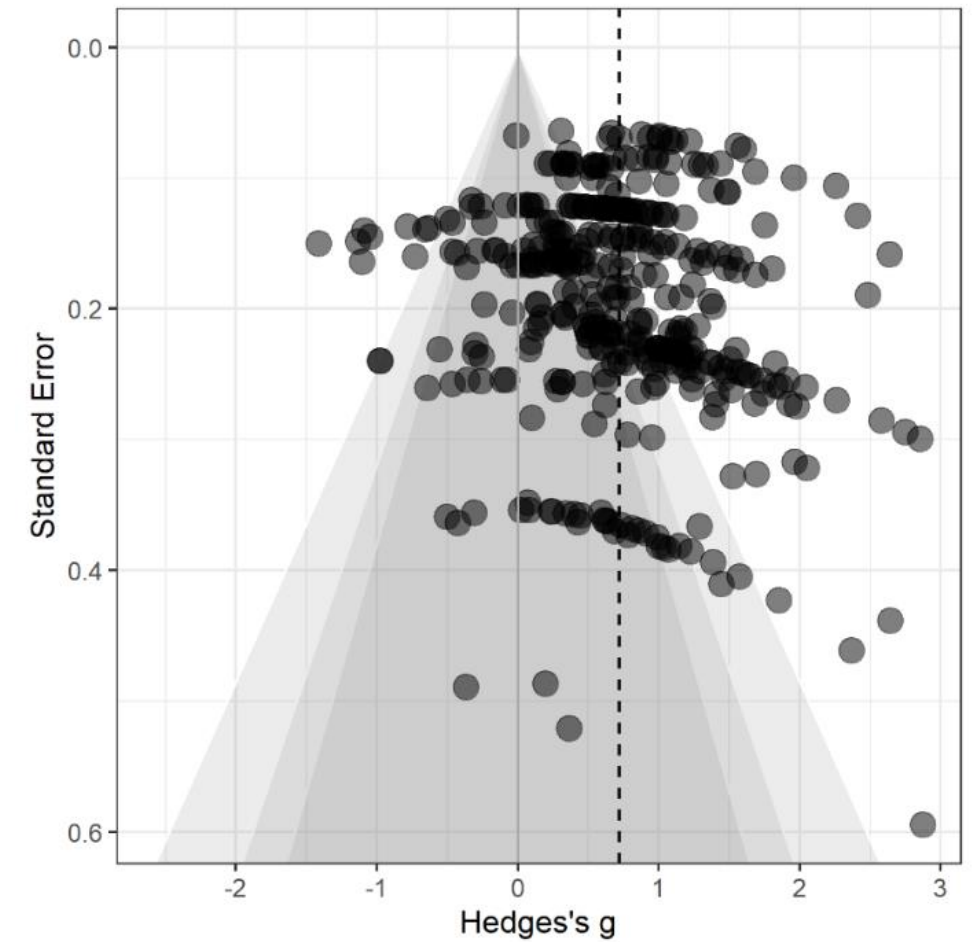
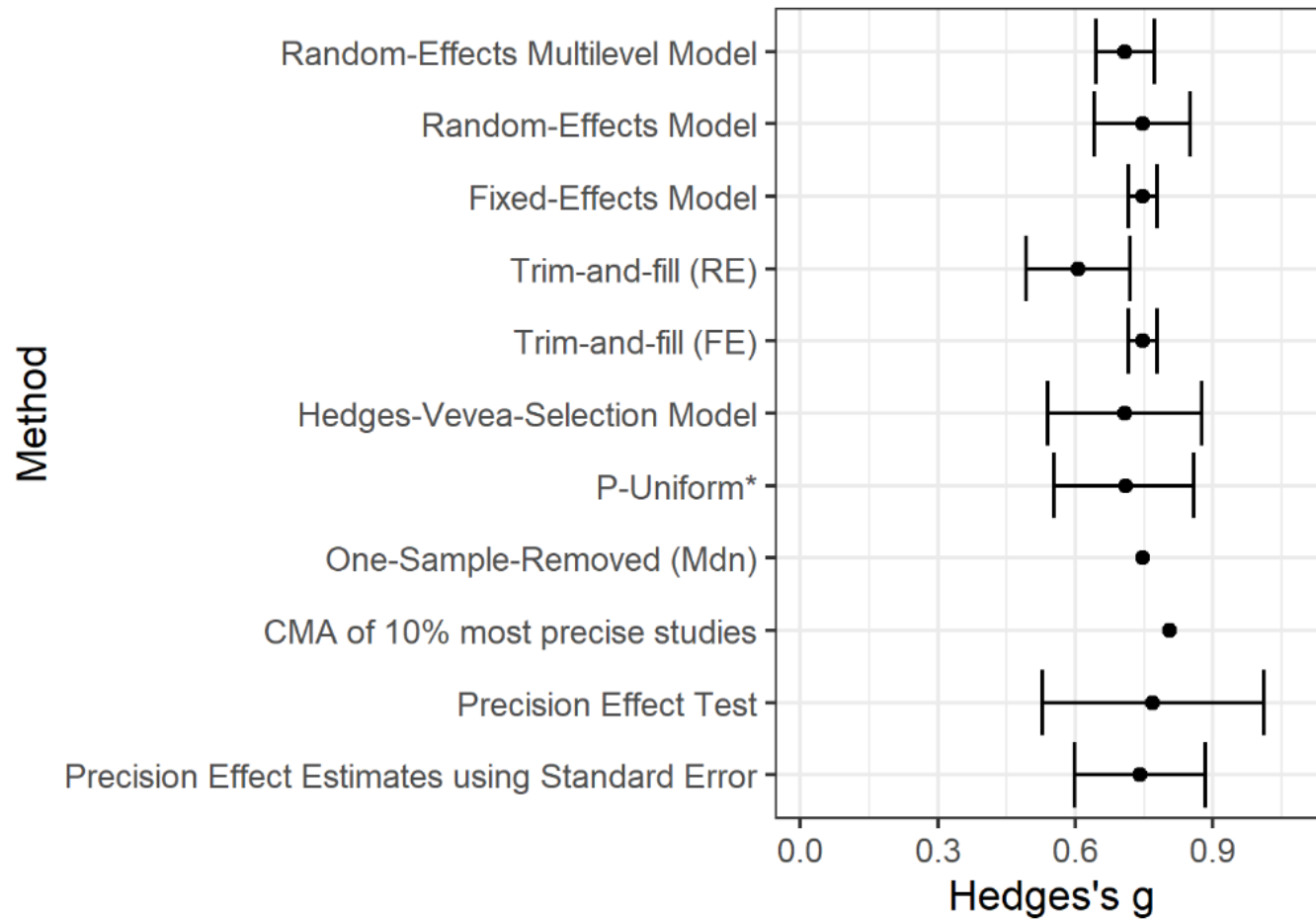
We have been accumulating trial-level data for all open anchoring datasets and reaching out to authors for their original data

This is what this looks like



participant_id	sex	age	anchoring_item	true_value	anchor	anchorhigh	estimate	tasktype	direction	incentive	comparative_question	experiment_type	
1001			Length of the Rio Japura (km)	2816	1408	0	5000	length	1	0	0	1	
1001			Length of the Lena (km)	4294	2147	0	3500	length	1	0	0	1	
1001			Length of the Vistula (km)	1047	524	0	1000	length	1	0	0	1	
1001			Length of the Po (km)	652	326	0	560	length	1	0	0	1	
1001			Length of the Ems (km)	371	186	0	200	length	1	0	0	22	1

Data Quality



Data Quality

Moderators

Pre-Registered ▼

Show 10 ▼ entries

Search:

Moderator_Levels	g	ci_lower	ci_upper	k
no	0.702	0.577	0.826	206
yes	0.754	0.585	0.924	199

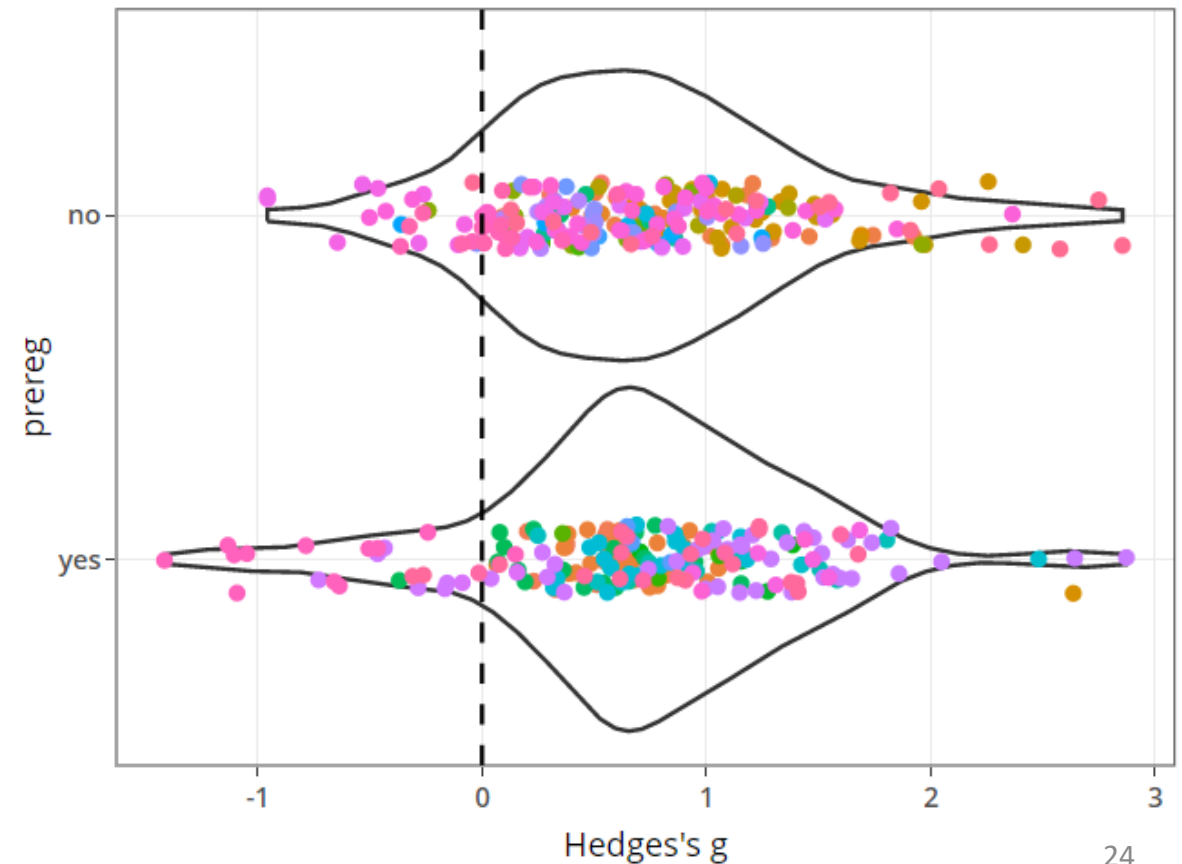
Showing 1 to 2 of 2 entries

Previous

1

Next

The effect of **prereg** on anchoring effect sizes is **not** significant at the 5% level. Test of moderators: $F(1, 403) = 0.24$, $p = 0.622$.



Moderator	Past findings	This MA
Incentives for accurate estimates	- <i>Incentives decrease deliberate adjustment: Epley & Gilovich, 2005; LeBoeuf & Shafir, 2009; Meub et al., 2013</i> - No difference: Enke et al., 2021; L. Li et al., 2021; Simmons et al., 2010; Tversky & Kahneman, 1974; Wilson et al., 1996	No difference
Age	—	No relationship
Gender	—	No relationship
Adjustment direction	—	No difference
Lab vs. online experiments	- <i>Larger variance in nonlab experiment: L. Li et al., 2021, Table 3</i>	No difference
Anchor extremeness	- Nonlinearity: Chapman & Johnson, 1994; Thorsteinson, 2011 - <i>Possible inverted u-shape: Bystranowski et al., 2021, p. 13; Townson, 2019</i>	Positive and nonlinear relationship between extremeness and effect size
Description of stimulus	- <i>Audio/video > brief written description > detailed written description; Bystranowski et al., 2021</i>	Image/audio/video < brief written description < other
Comparative question	—	Larger effects when a comparative question was present (probably due to incidental anchors not having CQ)
Scale type	- <i>No difference: Bystranowski et al., 2021, p. 13</i>	Open > visual > closed
Anchor manipulation (continuous vs. high-low)	- No difference: Bystranowski et al., 2021, p. 14	No difference
Type of anchoring task (estimating weight, height, price, etc.)	—	predictions > quantity > length > temperature > height > date > price [considering only k > 10 studies]
Relevance of the anchor	- Smaller effects for irrelevant anchors: L. Li et al., 2021, Table 3; Frederick et al., 2014; Schweickart et al., 2020; Sugden et al., 2013; Townson, 2019 - <i>No difference: English et al., 2006, p. 194; Bystranowski et al., 2021, p. 14</i>	Relevant > fixed > random

Overview of Anchoring Moderators

Probably

- Precision
- Distance (in a non-linear way)
- Relevance of anchor (e.g., random or not)
- Description of Stimulus
- Type of Response Scale

Probably not

- Incentives/Forewarnings
- Personality / Cognitive Ability
- Cognitive Bias
- Time-pressure
- Ego Depletion
- Subliminal Anchors
- Incidental Anchors
- Knowledge about Anchoring

The Future of Anchoring

- With the OpAQ, anchoring can call itself a poster child for Open Science
- Rigorous research has shown that the prominent models of anchoring are wrong
- Creating new models has been difficult due to all sorts of contradictions
- We are finally in a position to derive new theories



Core Team



Lukas Röseler,
PhD



Lucia Weber,
BSc



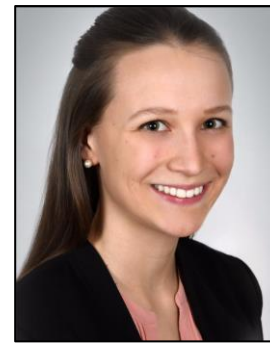
Katharina
Helgerth



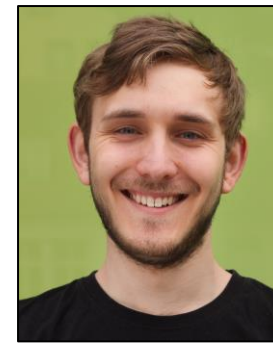
Elena Stich



Miriam
Günther



Paulina
Tegethoff, BSc



Felix Wagner



Astrid Schütz,
PhD

Resources Contributors

Röseler, L., Weber, L., Stich, E., Günther, M., Helgerth, K. A. C., Tegethoff, P., Wagner, F. S., Antunovic, M., Bahník, S., Barrera, F., Baumeister, R. F., Bermeitinger, C., Bickenbach, S. L. C., Blank, P. A., Blower, F. B. N., Böglér, H. L., Boo, F. L., Boruchowicz, C., Bühler, R. L., Burgmer, P., Cheek, N., N., Dohle, S., Dorsch, L., Dück, M. S., Halali, E., Fels, S.-A., Fischer, A. L., Frech, M.-L., Freira, L., Friedinger, K., Genschow, O., Harris, A., Häusser, J. A., Henkel, M., Horvath, D., Hügel, J. C., Igna, E. L. E., Imhoff, R., Intelmann, P., Ioannidis, K., Karg, A. H., Klamar, A., Klusmann, B., Knappe, E., Köppel, L.-M., Koßmann, L., Kraft, P., Kroworsch, M. K., Krueger, S. M., Kühling, S., Lammers, J., Loschelder, D. D., Milstein, N., Molden, D. C., Navajas, J., Norem, J., K., Novak, J. Page, E., Panse, F., Papenmeier, F., Pavlovic, Z., Rebholz, T. R., Rinn, R., Rodgers, S., Röseler, J. J., Roßmaier, K. V., Sartorio, M., Scheelje, L., Schindler, S., Schreiner, N. B., Schreiter, M. L., Seida, C., Shanks, D. R., Siems, M.-C., Stitz, M., Starkulla, M., Stäglich, M., Thies, K., Thum, E., Undorf, M., Unger, B. D., Urlichich, D., Vadillo, M. A., Wackershauser-Sablotny, V., Wagner, F. S., Wessel, I., Wolf, H., Zhou, A., Zimdahl, M.



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